Confluent Platform and Apache Flink on Minikube

A Makefile-driven quickstart that deploys a full local streaming stack on Minikube:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.2** via the Flink Kubernetes Operator
- **Kafka UI** (Provectus) for cluster inspection

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1.0 Prerequisites

macOS with Homebrew. To install all required tools in one step:

```
make install-prereqs
```

This installs Docker Desktop, `kubectl`, and Minikube via Homebrew. Once complete, **launch Docker Desktop** before proceeding.

To verify all tools are present without installing:

```
make check-prereqs
```

Required: `docker`, `kubectl`, `minikube`, `helm`, `envsubst` (`brew install gettext`).

2.0 Resource Requirements

Minikube is configured with the following defaults, which are required to run the full stack:

Resource	Default
CPUs	6
Memory	20 GB
Disk	50 GB

Override any of these at the command line:

```
make up MINIKUBE_CPUS=8 MINIKUBE_MEM=24576
```

3.0 Architecture

```
graph TD
    %% ── Composite entry points ──────────────────────────────────
    UP(["`**make up**`"])
    FLINK_UP(["`**make flink-up**`"])
    DOWN(["`**make down**`"])
    FLINK_DOWN(["`**make flink-down**`"])
    TEARDOWN(["`**make teardown**`"])

    %% ── Phase 1: Prerequisites ──────────────────────────────────
    subgraph P1["Phase 1 — Prerequisites"]
        CHECK_PRE["check-prereqs\ndocker · kubectl · minikube · helm"]
    end

    %% ── Phase 2: Minikube ───────────────────────────────────────
    subgraph P2["Phase 2 — Minikube"]
        MK_START["minikube-start\ncpus=6 · mem=20GB · disk=50GB"]
        MK_STOP["minikube-stop"]
    end

    %% ── Phase 3: Confluent Operator ─────────────────────────────
    subgraph P3["Phase 3 — Confluent Operator"]
        NS["namespace\nkubectl create namespace confluent"]
        OP_INSTALL["operator-install\nhelm: confluent-for-kubernetes"]
        OP_UNINSTALL["operator-uninstall"]
```

```
    end

    %% ── Phase 4: Confluent Platform
────────────────────────────────────────
    subgraph P4["Phase 4 — Confluent Platform"]
        CP_DEPLOY["platform-deploy\nKafka KRaft · SR · Connect\nksqlDB ·
REST Proxy · C3"]
        CP_DELETE["platform-delete"]
    end

    %% ── Phase 5: Control Center
──────────────────────────────────────────
    subgraph P5["Phase 5 — Control Center"]
        C3["c3-open\nlocalhost:9021"]
    end

    %% ── Phase 6: Apache Flink
─────────────────────────────────────────────
    subgraph P6["Phase 6 — Apache Flink"]
        CERT["flink-cert-manager\ncert-manager v1.17.1"]
        FL_OP["flink-operator-install\nhelm: flink-kubernetes-operator
1.14.0"]
        FL_DEPLOY["flink-deploy\nenvsubst → FlinkDeployment
CR\nflink:2.2"]
        FL_UI["flink-ui\nlocalhost:8081"]
        FL_DELETE["flink-delete"]
        FL_OP_UN["flink-operator-uninstall"]
        CERT_UN["cert-manager-uninstall"]
    end

    %% ── Phase 7: Kafka UI
──────────────────────────────────────────────
    subgraph P7["Phase 7 — Kafka UI"]
        KUI_INSTALL["kafka-ui-install\nhelm: provectus/kafka-
ui\nbootstrap: kafka:9092"]
        KUI_OPEN["kafka-ui-open\nlocalhost:8080"]
        KUI_UN["kafka-ui-uninstall"]
    end

    %% ── Manifests / Templates
─────────────────────────────────────────────
    subgraph FS["k8s/base/"]
        MANIFEST[("flink-basic-deployment.yaml\nFLINK_IMAGE ·
FLINK_VERSION")]
    end

    %% ── make up dependency chain
──────────────────────────────────────────
    UP --> CHECK_PRE
    UP --> MK_START
    UP --> CP_CORE_UP
    UP --> KUI_INSTALL

    CP_CORE_UP["cp-core-up"] --> OP_INSTALL
```

```
    CP_CORE_UP --> CP_DEPLOY
    OP_INSTALL --> NS

    %% ── make flink-up dependency chain
────────────────────────────────────
    FLINK_UP --> CERT
    FLINK_UP --> FL_OP
    FLINK_UP --> FL_DEPLOY
    FL_OP --> NS
    FL_DEPLOY --> MANIFEST

    %% ── make down dependency chain
──────────────────────────────────────
    DOWN --> KUI_UN
    DOWN --> CP_DELETE
    DOWN --> OP_UNINSTALL

    %% ── make flink-down dependency chain
────────────────────────────────────
    FLINK_DOWN --> FL_DELETE
    FLINK_DOWN --> FL_OP_UN
    FLINK_DOWN --> CERT_UN

    %% ── make teardown
────────────────────────────────────────────────────────
    TEARDOWN -->|"1 — check minikube running"| FLINK_DOWN
    TEARDOWN -->|"2"| DOWN
    TEARDOWN -->|"3 — delete namespace"| NS
    TEARDOWN -->|"4"| MK_STOP

    %% ── UI access
────────────────────────────────────────────────────────────
    CP_DEPLOY -.->|"once Running"| C3
    FL_DEPLOY -.->|"once Running"| FL_UI
    KUI_INSTALL -.->|"once Running"| KUI_OPEN

    %% ── Styles
──────────────────────────────────────────────────────────────
    classDef entry    fill:#1a1a2e,stroke:#e94560,color:#fff,font-
weight:bold
    classDef install  fill:#16213e,stroke:#0f3460,color:#a8dadc
    classDef remove   fill:#2d1b1b,stroke:#8b0000,color:#ffb3b3
    classDef ui       fill:#1b2d1b,stroke:#2d6a2d,color:#b3ffb3
    classDef file     fill:#2d2b1b,stroke:#8b7500,color:#ffe680
    classDef composite fill:#2a1a2e,stroke:#9b59b6,color:#dbb8ff

    class UP,FLINK_UP,DOWN,FLINK_DOWN,TEARDOWN entry
    class CP_CORE_UP composite
    class
CHECK_PRE,MK_START,NS,OP_INSTALL,CP_DEPLOY,CERT,FL_OP,FL_DEPLOY,KUI_INSTAL
L install
    class MK_STOP,OP_UNINSTALL,CP_DELETE,FL_DELETE,FL_OP_UN,CERT_UN,KUI_UN
remove
```

```
    class C3,FL_UI,KUI_OPEN ui
    class MANIFEST file
```

4.0 Quickstart

4.1 Full stack (CP + Kafka UI)

```
make up
```

This runs: `check-prereqs` → `minikube-start` → `namespace` → `operator-install` → `platform-deploy` → `kafka-ui-install`.

Once pods are up, open Control Center:

```
make c3-open        # http://localhost:9021
```

4.2 Add Apache Flink (run separately after `make up`)

```
make flink-up
```

This runs: `namespace` → `flink-cert-manager` → `flink-operator-install` → `flink-deploy`. `flink-up` is self-contained and can also be run standalone on a fresh cluster.

Once the Flink JobManager pod is running:

```
make flink-ui       # http://localhost:8081
```

5.0 Composite Workflow Reference

Target	What it does
`make up`	Full stack: Minikube + CP + Kafka UI
`make flink-up`	cert-manager + Flink Operator + Flink cluster
`make down`	Remove CP, Kafka UI, and Operator (Minikube keeps running)
`make flink-down`	Remove Flink cluster, Operator, and cert-manager
`make teardown`	Full teardown: everything + stop Minikube

6.0 Individual Target Reference

6.1 Phase 1 — Prerequisites

Target	Description
`install-prereqs`	Install Docker Desktop, kubectl, Minikube via Homebrew
`check-prereqs`	Verify all required tools are available

6.2 Phase 2 — Minikube

Target	Description
`minikube-start`	Start Minikube with configured resources
`minikube-status`	Show Minikube and node status
`minikube-stop`	Stop the Minikube cluster
`minikube-delete`	Permanently delete the Minikube cluster

6.3 Phase 3 — Confluent Operator

Target	Description
`namespace`	Create the `confluent` namespace and set it as default context
`operator-install`	Add Confluent Helm repo and install CFK Operator
`operator-status`	Show CFK Operator pod status
`operator-uninstall`	Remove the CFK Operator Helm release

6.4 Phase 4 — Confluent Platform

Target	Description
`platform-deploy`	Deploy Kafka (KRaft), Schema Registry, Connect, ksqlDB, REST Proxy, Control Center
`platform-watch`	Watch pod startup live (Ctrl+C to exit)
`platform-status`	Show current pod status
`platform-delete`	Remove all CP components

6.5 Phase 5 — Control Center

Target	Description
`c3-open`	Port-forward Control Center and open `http://localhost:9021`

6.6 Phase 6 — Apache Flink

Target	Description
`flink-cert-manager`	Install cert-manager (Flink Operator dependency)
`flink-operator-install`	Install the Flink Kubernetes Operator
`flink-operator-status`	Show Flink Operator pod status
`flink-operator-uninstall`	Remove the Flink Operator Helm release
`flink-deploy`	Deploy the Flink session cluster
`flink-status`	Show Flink pods and FlinkDeployment CRs
`flink-ui`	Port-forward Flink UI and open `http://localhost:8081`
`flink-delete`	Delete the Flink session cluster
`cert-manager-uninstall`	Remove cert-manager

6.7 Phase 7 — Kafka UI

Target	Description
`kafka-ui-install`	Install Kafka UI connected to the local CP cluster
`kafka-ui-status`	Show Kafka UI pod status
`kafka-ui-open`	Port-forward Kafka UI and open `http://localhost:8080`
`kafka-ui-uninstall`	Remove Kafka UI

7.0 Configuration

All variables are overridable at the command line. Defaults:

Variable	Default	Description
`NAMESPACE`	`confluent`	Kubernetes namespace
`MINIKUBE_CPUS`	`6`	vCPUs allocated to Minikube
`MINIKUBE_MEM`	`20480`	Memory in MB
`MINIKUBE_DISK`	`50g`	Disk size
`FLINK_OPERATOR_VER`	`1.14.0`	Flink Kubernetes Operator version
`FLINK_IMAGE`	`flink:2.2`	Flink container image
`FLINK_VERSION`	`v2_2`	Flink API version string for the FlinkDeployment CR

Variable	Default	Description
`FLINK_CLUSTER_NAME`	`flink-basic`	Name of the FlinkDeployment resource
`FLINK_MANIFEST`	`k8s/base/flink-basic-deployment.yaml`	Path to FlinkDeployment template
`CERT_MANAGER_VER`	`v1.17.1`	cert-manager version
`C3_PORT`	`9021`	Control Center local port
`FLINK_UI_PORT`	`8081`	Flink UI local port
`KAFKA_UI_PORT`	`8080`	Kafka UI local port

Example — deploy Flink 2.1 instead of 2.2:

```
make flink-deploy FLINK_IMAGE=flink:2.1 FLINK_VERSION=v2_1
```

8.0 Repository Layout

```
.
├── Makefile
├── README.md
├── README.pdf
├── CHANGELOG.md
├── CHANGELOG.pdf
├── KNOWN_ISSUES.md
├── KNOWN_ISSUES.pdf
├── .gitignore
└── docs
    └── manual_deployment.md   # Step-by-step manual deployment
instructions (without Makefile)
    └── manual_deployment.pdf  # Step-by-step manual deployment
instructions (without Makefile)
└── k8s/
    └── base/
        └── flink-basic-deployment.yaml   # FlinkDeployment CR template
```

The `flink-basic-deployment.yaml` is a template — `FLINK_IMAGE` and `FLINK_VERSION` are substituted at deploy time via `envsubst`. Do not apply it directly with `kubectl apply`.

9.0 Teardown

Remove everything and stop Minikube:

```
make teardown
```

To keep Minikube running but remove all deployed components:

```
make flink-down   # Flink cluster + operator + cert-manager
make down         # CP + Kafka UI + CFK Operator
```

10.0 Manual Deployment Instructions

For users who want to understand the underlying steps without using the Makefile, see docs/manual_deployment.md.